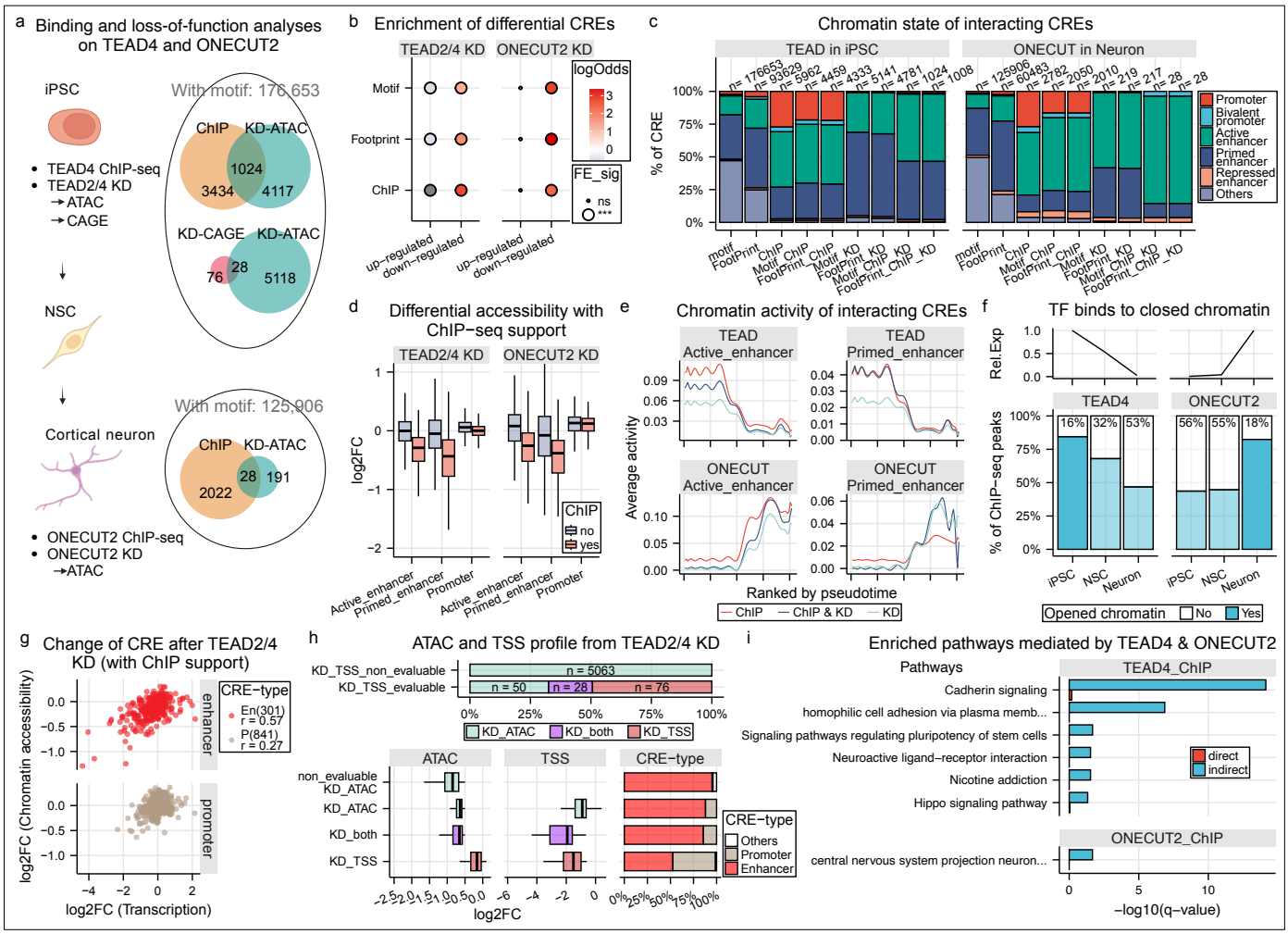


Figure 4 | TEAD4 and ONECUT2 preferentially regulate chromatin accessibility of enhancers.

a, Schematic summarizing the number of CREs associated with TEAD4 and ONECUT2 in iPSC and Neuron respectively. **b**, Enrichment of differentially accessible CREs among CREs containing corresponding TF motifs, showing positive TF footprinting, or supported by TF binding identified through ChIP-seq. **c**, Chromatin state distribution of CREs associated with TEAD and ONECUT. **d**, Overall log2 fold change of differential accessibility after knockdown of TEAD2/4 and ONECUT2. Distributions of log2 fold changes of CREs of different chromatin states with and without ChIP-seq support are shown. **e**, Upper: Mean chromatin accessibility of TEAD-associated CREs across the differentiation pseudotime. CREs with motif positive were stratified by ChIP-seq support, significantly reduced accessibility or both. Accessibility profiles are shown separately for active and primed enhancers. Lower: corresponding analysis for ONECUT-associated CREs. **f**, Overlap of TEAD4 and ONECUT2 ChIP-seq peaks containing their respective motifs and ATAC-seq peaks identified in the 3 samples. Percentages indicate the proportion of ChIP-seq peaks without overlap with accessible chromatin regions in the corresponding cell type. The upper panel shows relative TF expression based on mean expression levels across cell types. **g**, Correlation between changes in chromatin accessibility (ATAC-seq) and transcription (CAGE) following TEAD2/4 knockdown in iPSCs, measured as log2 fold changes across promoters and enhancers with ChIP-seq support and evaluable in both differential activity analyses. **h**, TEAD motif-containing CREs showing significant down-regulation of either ATAC or TSS signal upon TEAD2/4 KD, stratified by CAGE evaluability and differential response in the two modalities. Distribution of ATAC-seq and CAGE log2 fold changes and proportion of CRE-types in iPSC are shown for the corresponding CRE groups. **i**, Pathway enrichment of genes directly and indirectly associated with TEAD4 in iPSC and ONECUT2 in neuron, based on CREs supported by ChIP-seq. For indirect associations, expressing genes linked to TEAD-bound enhancers according to ABC model (ABC score > 0.02) were included. Gene Ontology (GO) enrichment was performed using topGO with the elim algorithm, while KEGG and Reactome pathway enrichment analyses were performed using clusterProfiler. Only significant pathways (q-value < 0.05) are shown.